Detecting, quantifying, and attributing fundamental disagreements across annotator groups

Anonymous ACL submission

Abstract

Current annotation agreement metrics are often sensitive to group size imbalances and noise, making them inadequate for many subjective tasks such as toxicity and hate-speech detection, where minority opinions are most important. Polarization has recently been proposed as a more robust way to distinguish minor disagreements from systematic differences in opinion, but existing approaches do not provide practical tools for attributing polarization to specific annotator groups. We evaluate current methods and identify two major limitations in realistic settings: (1) the presence of "inherent" polarization that cannot be attributed to any known or latent groups, and (2) opposing polarization effects canceling each other out in aggregated annotations. To address these issues, we introduce a new metric that measures and tests the statistical significance of polarization attribution for annotator groups while avoiding these limitations. We apply our method to four subjective NLP datasets and find that gender and race consistently explain polarization patterns, while differences between annotator groups become stronger as the groups are further apart. Finally, we estimate the minimum number of annotators required for reliable results and release an open-source Python library implementing our approach.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity, and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable (United Nations, 2025; Dioum, 2025). However, annotations for such tasks are subjective and typically aggregated based on majority-voting (Ivey et al., 2025; Giannoutsos et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025). This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems (Sap et al., 2019; van der Velden et al., 2025).

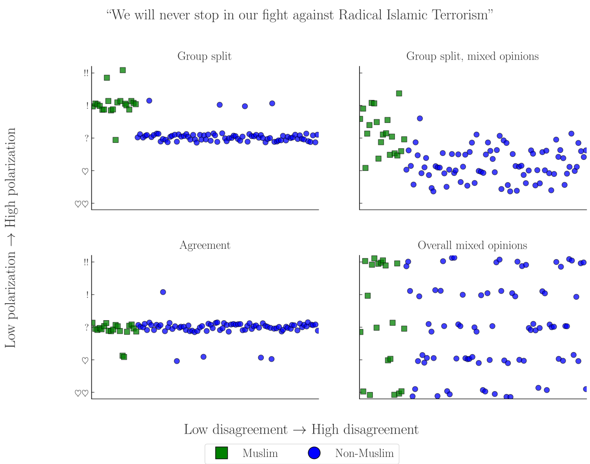

Figure 1: Simulated experiment exploring four scenarios of a hate speech detection task applied to a controversial statement by D. Trump (Jan. 2019). The variance in annotations across different annotators–Muslim annotators in the upper plots–may cause traditional agreement metrics may fail to capture distinct group biases (Pavlopoulos and Likas, 2024), especially when mixed opinions occur (top-right).

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These *fundamental* disagreements are typically ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). For example, coded language may not be picked up by most annotators but only by ones belonging in the targeted minority (Quaranto, 2022), as shown in Fig. 1. Furthermore, majority-group annotators may themselves be biased against minority groups, e.g., toxicity models disproportionally marking African American (AA) speech as "toxic" (Sap et al., 2019).

Instead of using inter-annotator agreement, we could use *polarization*, which formulates annotation analysis as a cluster identification problem (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization (Pavlopoulos and Likas, 2024), there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question:

RQ. *Can we attribute polarization to annotator subgroups?*

We first identify limitations of current work on polarization:

1. Existing mechanisms (Pavlopoulos and Likas, 2024) operate only as binary classifiers, producing non-quantifiable/comparable results.
2. These mechanisms only test for the extreme case where *all* present polarization can be attributed to a single annotator group, which we find to be extremely rare in real-life datasets (App. B).
3. We are the first to report the existence of *inherent polarization*: most comments having some degree of polarization that cannot be attributed to *any possible partition* of annotators (Finding 1).
4. We also find that annotator groups may exhibit opposing polarization attribution between comments, requiring isolating annotations to each individual comment when analyzing whole datasets (Finding 2).

We then introduce a new metric ("Aposteriori Unimodality–*apunim*") that attributes polarization to specific annotator groups without and develop a parametric test for statistical significance. Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables *quantitative* comparisons across datasets. By considering the entire set of comments in a dataset *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets.

Having established our metric, we apply it to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection (§5), finding that annotator gender and race significantly impacts annotation (Finding 3), and that fundamental disagreements grow more pronounced the further apart annotator groups are (e.g., irreligious vs. very religious annotators–Finding 4).

Finally, we release an open source Python library that implements our metric, which can be directly installed via PyPi.[1] and provide an estimation on the minimum number of annotators required for robust estimates (§6). The code for the experiments, graph creation, and analysis can be found on our project's repository.[2]

This publication includes examples of controversial and potentially harmful speech for the purposes of demonstration. These examples do not represent the views of the authors.

2 Related Work

2.1 Agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen's kappa, Krippendorff's alpha and Fleiss's kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or "random disagreement") using statistical methods. However, most cannot be generally used for comparing intergroup annotation agreement without extending them (e.g., computing pairwise Cohen's kappa for all groups), or placing constraints on when they can be used (e.g., no missing labels and same number of annotations per annotator for Fleiss's kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), for their inability to quantify and compare results across experiments (Wong et al., 2021), and for fundamentally faulty assumptions around the concept of "chance-correction" (Checco et al., 2017).

Recent work has attempted to work around these limitations by using simpler statistical models (van der Velden et al., 2025), adapting Cohen's kappa for subgroup agreement (Wong et al., 2021), or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise (Ivey et al., 2025). While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs outgroup variance, but still necessitates traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric polarization metric, normalized Distance From Unimodal-

[1]Library: `https://anonymous.4open.science/r/apunim-5F70/README.md`
Documentation: `https://anonymous.4open.science/r/apunim-page/`

[2]Replication code: `https://anonymous.4open.science/r/aposteriori-unimodality-E8C1/`

ity (nDFU), which has been shown to correlate better with human judgment of disagreement than established metrics (Pavlopoulos and Likas, 2024). nDFU takes values in the range [0, 1], where values close to 0 indicate little or no polarization (i.e., opinions are concentrated around a single mode), while values close to 1 indicate strong polarization with multiple distinct peaks. The metric is defined as follows:

$$\mathrm{nDFU}(X) = \frac{\max\limits_{i \neq m}\left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases}\right)}{f_m} \qquad (1)$$
$$m = \arg\max_i f_i$$

where, X denotes the set of annotations, f_i the relative frequency of the i-th annotation, and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak.

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing polarization to subgroups is a relatively new idea. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are "causing" the exhibited polarization. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split. Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.[3]

3 What are the limitations of polarization attribution?

3.1 Problem Formulation

Let $d = \{c_1, c_2, \ldots\}$ be a dataset composed of multiple annotated comments.[4] Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \ldots\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender,

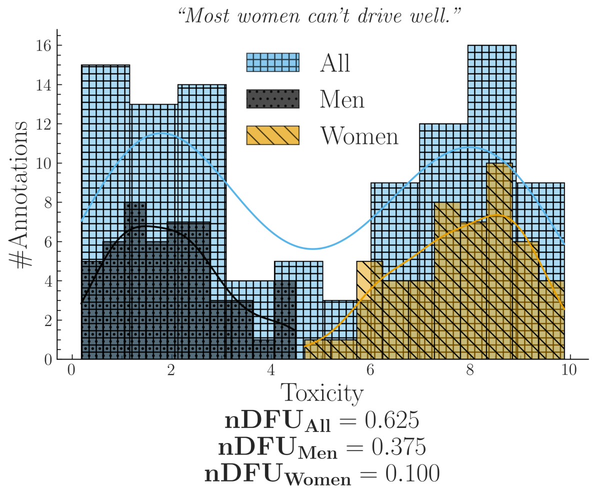

Figure 2: Synthetic experiment simulating a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

would be formulated as:

$$A(\text{"Could be better, could be worse"}) \\ = \{\ (+, F), (-, M), (+, F), \ldots\ \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c \mid g) = \{a(c; g') \in A(c) \mid g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Consider the (simulated) case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2. The annotations are generally polarized ($nDFU_{all} = 0.62$); but both female ($nDFU_{women} = 0.10$) and male ($nDFU_{men} = 0.37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is driven by substantive disagreements between male and female annotators.

3.2 Inherent polarization

Inherent polarization refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024) attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU(all) > nDFU(men) = nDFU(women) = 0$). The presence of inherent polarization would invalidate this theory, since there would be no possible group g s.t. $nDFU(g) = 0$.

Definition 1: Inherent polarization. *Polarization that cannot be explained by any annotator known or latent annotatot grouping.*

[3] For example, when presented with noisy estimates of a metric in the [0, 1] range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

[4] These items could be of any modality, such as comments in a discussion, images, videos or survey questions, even though we consider comments in this study.

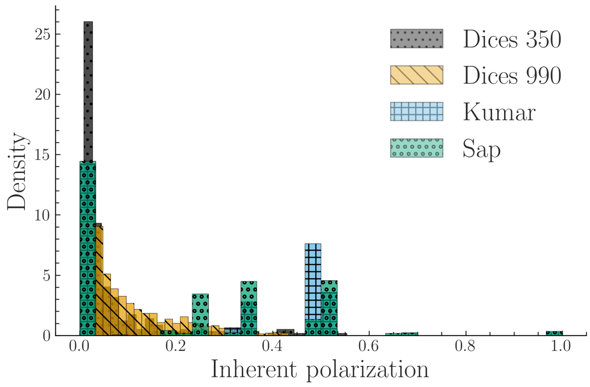

Figure 3: Apriori (inherent rather than observed) polarization across our datasets. Contrary to assumptions made in prior work, inherent polarization is rarely zero and can even reach $nDFU = 1$.

Formally, the inherent polarization for a single comment c is given by:

$$Pol_{inh}(c) = \min_i \{nDFU(\tilde{r}_i(A(c)))\} \quad (2)$$

where $\tilde{r}$ is the random partition operator.

To examine this assumption, we create random annotator groupings for each comment in our datasets. Specifically, for every comment, we randomly partition the annotators into random groups (with randomly selected sizes) 100 times and compute the observed polarization for each partition. We then take the *minimum* observed polarization scores across all random partitions, which guarantees that we check for all possible variables, known and latent. Repeating this process for every comment produces the distribution shown in Fig. 3, demonstrating that inherent polarization is consistently non-zero and can even be substantial for many comments.

Finding 1. *Each comment may exhibit a degree of inherent polarization that cannot be attributed to any known or latent annotator grouping.*

3.3 Conflicting polarization

Previous work (Pavlopoulos and Likas, 2024) operated on the comment level. In practice however, this approach is inherently noisy due to a lack of information (i.e., limited annotations) relating to each individual comment. Many datasets in NLP are annotated by only 3–5 annotators per comment, which is especially problematic when we want to analyze patterns in annotator subgroups; in a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split. This issue is especially prevalent for polarization, since *nDFU is undefined for groups of two* (and arguably marginally useful for groups of three), since it requires at least two histogram peaks to function (see Eq. 1), a limitation not acknowledged in the original work.

This limitation leaves us with two choices: (1) only use datasets featuring at least 6–8 annotators, or (2) aggregate annotations across comments for each dataset. The former case is unrealistic, especially for most NLP datasets due to annotation costs. We consider the latter case in an experiment shown in Fig. 4 where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might balance each other out, leading to a false negative. Therefore, we should never aggregate annotations across comments.[5]

Finding 2. *Polarization has a direction, which may compromise cross-comment annotation aggregation.*

4 How do we fix it?

4.1 From inherent to apriori polarization

In §3.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c \mid g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset–in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the nDFUs of men and women, which are similar as they are both unimodal distributions, falsely concluding that there is no polarization present.

We instead compare the observed polarization of $A(c)$ with its inherent polarization, following the methodology introduced in §3.2 with two key distinctions. (1) Instead of using random splits, we create random stratified splits.[6] (2) Instead of finding the minimum possible polarization exhibited by any known or latent group, we calculate the *expected* (average). Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*, which is a much stricter test.[7] Specifically, we define a-priori unimodality for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t} \sum_{i=1}^{t} nDFU(\tilde{r}_i(A(c))) \quad (3)$$

[5]A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

[6]I.e., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we create random partitions of sizes 80 and 20.

[7]Indeed, by definition, all groups exhibit equal or larger polarization than the inherent pol. value.

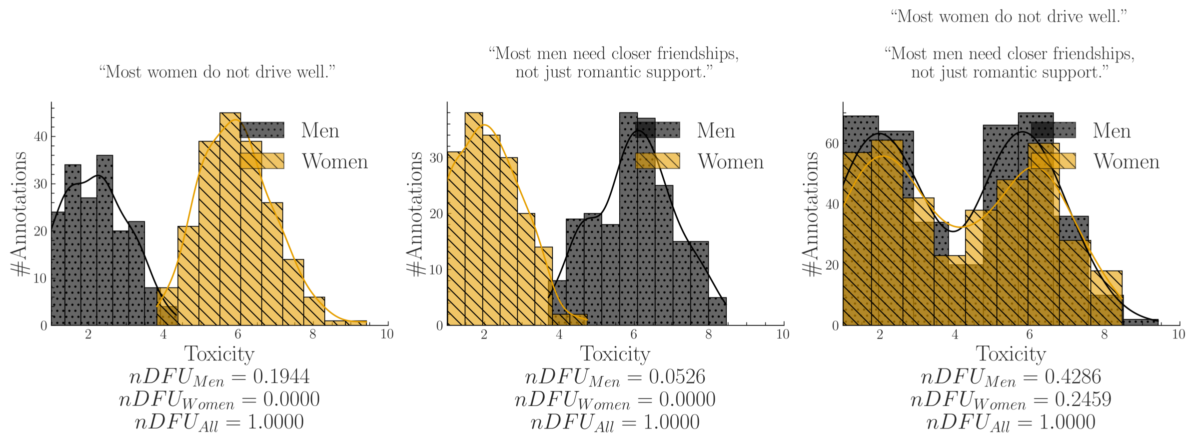

Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization.

Definition 2: A-priori polarization. *The expected polarization of a comment's annotations.*

4.2 Expanding to the whole dataset

As discussed in §3.3, we must not aggregate the annotations themselves before applying the nDFU metric. We can however estimate how much polarization is generally present in the dataset, by simply averaging the *a-priori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any group.

$$Pol^{d}_{apr} = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \tag{4}$$

Which we will compare with the average observed polarization of all comments:

$$Pol^{d}_{obs}(g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c \mid g)) \tag{5}$$

We can now compare the polarization that is attributed to a single-dimensional annotator group (Equation 5) to a prior one (Equation 4). We define our normalized metric, which we call *"apunim" (Aposteriori Unimodality)*, as:

$$apunim(d, g) = \frac{Pol^{d}_{obs}(g) - Pol^{d}_{apr}}{1 - Pol^{d}_{apr}} \tag{6}$$

Definition 3: Apunim. *The degree to which an annotator group contributes to overall polarization in a set of comments.*

Since $nDFU \to [0, 1] \Rightarrow apunim \to [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- $apunim(\theta) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(\theta) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(\theta) < 0$: Overall polarization *drops* when we add annotators from this group to this task. This occurs when intragroup polarization is higher than apriori polarization; indicating that annotators of that group are polarized between themselves.

Because *apunim* is fundamentally designed to quantify polarization, we exclude comments that exhibit no measurable polarization ($nDFU(c) = 0$). Furthermore, to enhance computational efficiency, we also exclude comments annotated exclusively by a single group (e.g., comments annotated only by male or only by female annotators when considering gender).

4.3 Statistical significance

Since annotation histograms may be noisy, we additionally assess whether the observed apunim value could arise by chance. To assess statistical significance, we construct a null distribution of apunim values from the randomized partitions and compare the observed apunim against this distribution using a one-sample Student-*t* test. Thus, the null hypothesis states that the observed polarization is indistinguishable from polarization induced by random annotator assignments:

$$H_0: \; apunim(d, g) = apunim_{\text{rand}}(d, g)$$

$$H_a: \; apunim(d, g) \neq apunim_{\text{rand}}(d, g)$$

where $apunim_{\text{rand}}(d, g)$ denotes the distribution of apunim values obtained from randomized annota-

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. *#Ann* refers to the number of annotations per comment (see App. B), and *#Dims* to sociodemographic dimensions (e.g., age, gender). Tasks are either toxicity (Tox) or hate speech detection (HS).

tor partitions. For a detailed explanation of the p-value computation and the assumptions underlying the test, refer to App. C.

5 Attributing real-life polarization to human groups

5.1 Datasets

We use four datasets from current NLP research that include sociodemographic information at the level of individual annotators; specifically, the "Kumar"(Kumar et al., 2021) and "Sap"(Sap et al., 2022) datasets, which are concerned with Toxicity and hate speech detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.[8] While the toxicity and hate speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

A brief overview of dataset characteristics is provided in Table 1. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct sociodemographic dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (Fig. 7).[9]

5.2 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided sociodemographic dimensions can partially explain polarization in the toxicity / hate speech annotations, as shown in Table 1.[10] Note that some groups are not included, either because the results were not statistically significant, or because there were no polarized items annotated by more than one annotator of these groups. See App. E for details on other parameters and execution times, and App. F for the full results.

Group	apunim	support
DICES-350		
Race=AfricanAmerican	-0.043	9077
Race=Asian	0.066	8138
DICES-990		
Age=GenX+	-0.046	14384
Age=GenZ	0.063	13741
Edu=High school and below	-0.075	7419
Gender=Man	0.028	32494
Gender=Woman	-0.025	33471
Race=AfricanAmerican	0.091	5810
Race=Latino	-0.133	6610
Race=Other	0.075	24321
Race=White	-0.120	11392
Sap		
Gender=Man	0.091	1439
Gender=Woman	-0.244	877
Kumar		
Race=AfricanAmerican	0.2072	356
ToxTarget=False	−0.0017	2199
ToxTarget=True	0.0329	1176
Parent=No	−0.0591	1868
Trans=Yes	0.5322	31
ReligionImportant=No	−0.1069	1069
SO=Bisexual	0.1639	283
ToxProblem=Never	0.2975	92

Table 2: Statistically significant ($p < 0.05$) aposteriori unimodality results across all four datasets. Gray rows show negative values. Consult App. F for the full results.

5.3 Which factors contribute to polarization?

Race Across all datasets apart from Sap, we observe a consistent pattern in which annotation polarization is linked to annotator ethnicity, and particularly by African American annotators. Additionally, *ethnicity is the strongest contributor to polarization*, with statistically significant effects for at least one major ethnic group in each dataset (AAs: −0.0428/Asians: 0.0659 for DICES-350, AAs: 0.2072 for Kumar–a close second, and all ethnicities for DICES-990). These results align with prior work suggesting that ethnic background is a systematic source of annotator disagreement rather

[8] For the latter, we consider only labels relating to racism (`Q3_bias_overall`), thereby transforming the task to hate speech detection.

[9] This systematic discrepancy could be attributed to the DICES datasets being composed of Large Language Model (LLM)-generated dialogues with already-aligned models tuned to avoid racist speech.

[10] We sample 20,000 comments from the Kumar dataset due to computational costs arising from the extraordinary amount of distinct participant groups.

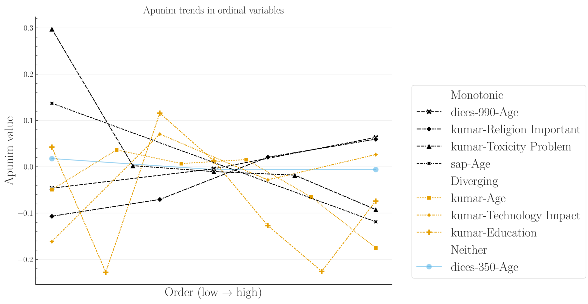

Figure 5: Polarization attribution trends in ordinal groups. The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and indifference towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in App. F.

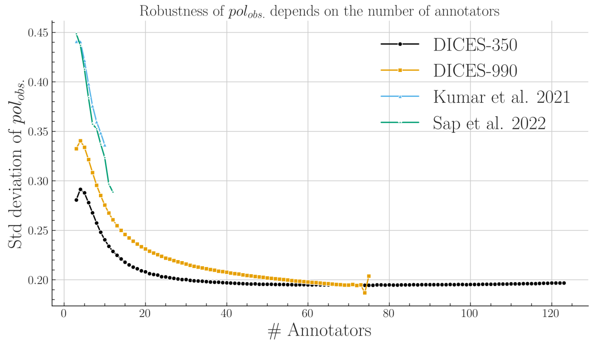

Figure 6: Standard deviation of 30 randomly sampled (with replacement) pol_{obs} values (Eq. 5). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (B).

than isolated noise (Goyal et al., 2022).

Gender Polarization consistently relates to gender across two of our hate speech datasets, with DICES-350 being the exception. DICES-990 shows that men and women have statistically significant, opposite effects (Men: 0.028; Women: −0.025). This pattern is similar to Sap, where men display a positive attribution score (0.091) and women display a strongly negative score (−0.244). These results are consistent with previous findings regarding hate speech (Wojatzki et al., 2018) and general toxicity annotation (Binns et al., 2017). Interestingly, the Kumar dataset–the only one providing this demographic data–indicates that transgender annotators showed the highest degree of polarization, even given their small representation in the dataset. This finding strongly supports our choice to use polarization as a metric, rather than depending on traditional inter-annotator agreement (Fig. 1).

Finding 3. *Annotator age and race are consistent causes of polarization, even when their representation in the annotations is sparse.*

Polarization attribution grows the more different annotator groups become Figure 5 shows *apunim* trends across ordinal sociodemographic attributes listed in Table 2. We find that polarization attribution changes consistently in one direction as we move from one end of an attribute scale to the other–for example, from irreligious to very religious annotators. In some sociodemographic dimensions this is expressed in a monotonic pattern, suggesting that perceived differences become stronger between groups that are further apart ideologically or demographically. In other cases, polarization attribution is caused by a few dominant modes.

Finding 4. *For ordinal sociodemographic dimensions, polarization attribution tends either to grow stronger as the differences between annotator groups increase, or coalesce around a few opposing groups.*

6 How many annotators do we need?

While *apunim* works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive (Rossi et al., 2024). Thus, it is imperative to find the least annotators needed for *apunim* to generate stable and reliable estimates. For each item per dataset, we randomly sample with replacement a subset of annotators and compute the standard deviation of 30 sampled pol_{obs} values (see Equation 5). Figure 6 demonstrates the effect of the number of annotators on the reliability of the polarization estimation, which we discuss next.

Each dataset features a different standard deviation even with the same number of annotators, which is broadly consistent with the amount of polarization in each dataset (App. B). However, we observe a clear, non-linear, inverse relationship between the number of annotators and the standard error across datasets. This can be attributed to a more robust histogram estimation used by nDFU (§4.1) when provided with more points for each of the subgroups. In the case of our datasets, we observe that after about 20 points, the estimation is good enough that more annotations contribute trivially to the robustness of the metric.

7 Conclusion & Future work

We investigated whether fundamental differences can be attributed to annotator groups using polarization, instead of standard agreement measures. We discovered as of yet unknown issues regarding the presence of inherent polarization and conflicting polarization directions which we bypass using a new metric, *apunim*. We then applied our proposed metric to four hate-speech and toxicity detection datasets, discovering that (1) race and gender seem to generally drive polarization attribution, and (2) groups that are further apart increasingly drive polarization. Furthermore, we estimated the robustness of our metric w.r.t. the number of annotators, and released a PyPi installable library implementing it.

Our framework has broad applications across NLP tasks related to content moderation. Because many of these tasks involve multiple discussions, the framework can also be applied at the discussion level rather than only at the dataset level. This would allow researchers to identify particularly polarizing discussions and study the underlying sources of disagreement for specific minority groups. Discussions could also be analyzed by comment turn, enabling the study of how polarization develops over the course of an interaction. Finally, our approach can be used to more reliably evaluate whether LLM-as-a-judge systems reproduce the polarization patterns observed among different minority groups.

8 Limitations

Researchers often release only aggregated single-label annotations or omit annotator sociodemographic characteristics entirely. As a result, both our analysis and the applicability of our framework are limited by the scarcity of suitable datasets. In addition, because datasets with large numbers of annotators per item are rare, our approach operates only at the dataset level. This limitation prevents us from identifying individual comments or examples that could serve as concrete explanations for the observed *apunim* values.

Finally, although we compare findings across datasets and relate them to prior work, we do not directly compare polarization and agreement metrics, since they rely on fundamentally different assumptions and mechanisms (see §2.2). These differences also make it difficult to fully explain certain patterns in our results, such as why some groups show statistically significant effects while others do not.

9 Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research. No identifiable, personal information is included in our pipeline. We did not perform any annotation tasks ourselves. Additionally, this study uses dichotomies across the two predominant genders, and religious groups as examples. The authors do not claim that these examples necessarily reflect the views or behaviors of these groups; they exist only for the purpose of demonstration. The controversial statements expressed in this publication also do not express the views of the authors.

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A Acronyms

LLM Large Language Model

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

B Annotation details

The number of annotators per comment for all datasets presented in §5.1 can be found in Figure 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 6 since they would produce uninterpretable figures.

C P-value estimation implementation

Since our test is parametric, it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per sociodemographic group to estimate nDFUs in Equations 3-5, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §5.1). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single group in principle, in practice we often want to test polarization for all subgroups of a specific dimension. Since we are simultaneously considering multiple hypotheses, we apply a multiple comparison correction to the resulting p-values–specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and apunim calculations, as well as its statistical test. The software is available on PyPi We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (`https://anonymous.4open.science/r/apunim-page/`).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project's repository (`https://anonymous.4open.science/r/aposteriori-unimodality-E8C1/`). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy, scipy, statsmodels`).

D.2 Implementation details

By default, the computation of the apunim values happens on a single thread. In cases of serious computational cost, we recommend running the apunim calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple sociodemographic dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{r}_i$), which are originally calculated to obtain the apriori polarization values (see Equation 3), for the p-value calculations (§4.3). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{r}_i$ in Equation 3), and the number of apunim values used in the p-value calculation have to be the same.

E Replication Details

We use version `1.0.1` of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 3) and the p-value estimation (§4.3), and set a FWER rate of 0.95.

While the library efficiently handles a large number of annotators through vectorized operations, performance degrades significantly when processing a high volume of comments, particularly when annotators belong to multiple distinct groups. For instance, while experiments across all datasets in §5.1 complete within a minute, the Kumar dataset necessitates approximately one hour of computation, even when using a 3% sample. Furthermore, the annotator variance analysis detailed in §6 required over ten hours. While this latter time investment is anticipated due to the inherently demanding nature and non-standard application of

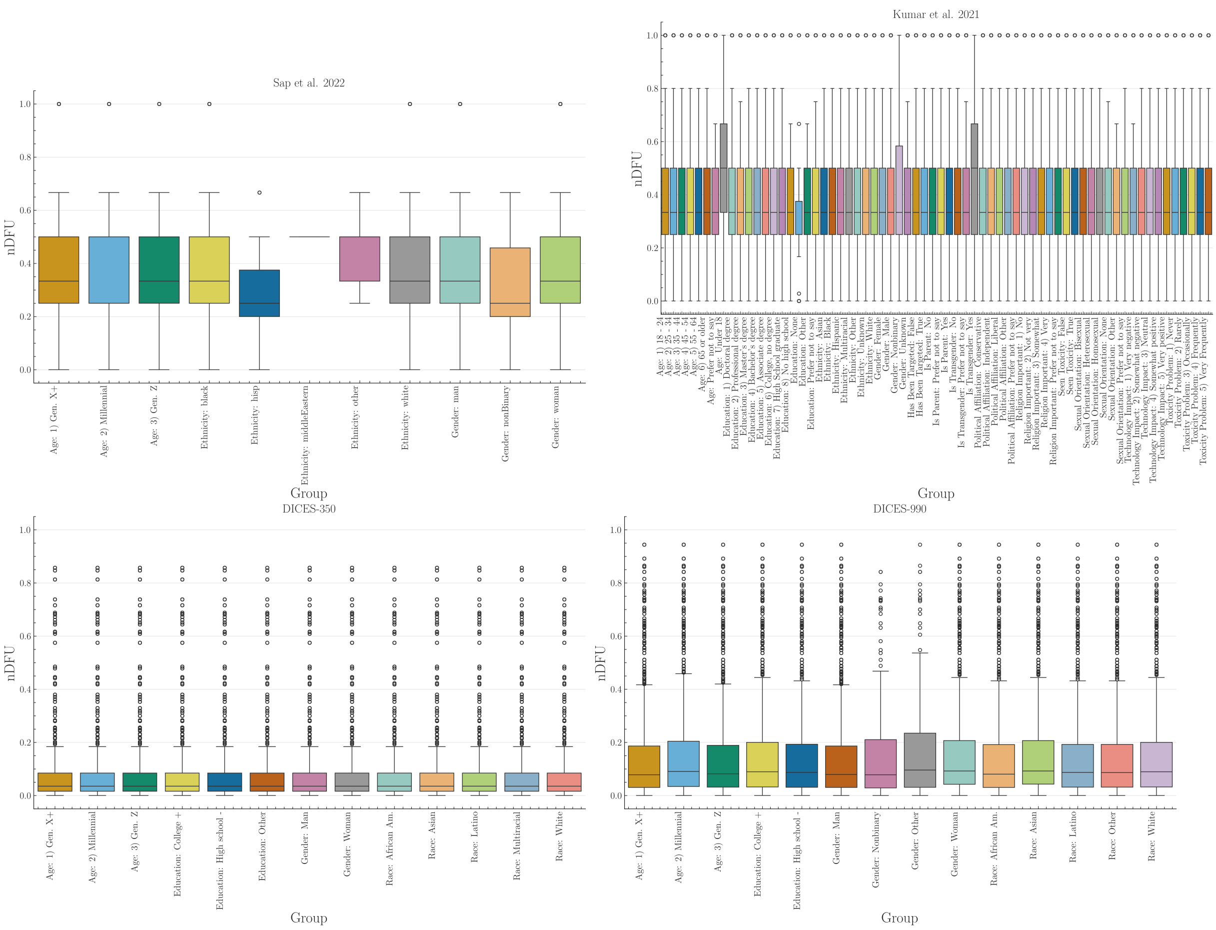

Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per sociodemographic dimension.

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

our library, it highlights a scaling bottleneck under specific data structures. We hypothesize that this bottleneck is likely caused by repeated creation of numpy arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each sociodemographic dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each sociodemographic subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package (Tange, 2025) for our own experiments).

In Figure 6, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see App. B).

F Full Results

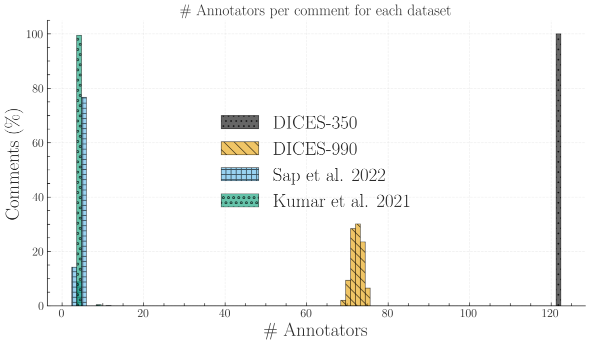

Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3).

Group	apunim	support
Sap		
Age=GenX+	0.0660	271
Age=Millennial	−0.0068	1886
Ethnicity=white	0.0004	1896
Gender=Man	0.0907***	1439
Gender=Woman	−0.2442***	877

Table 4: Aposteriori unimodality results for the Sap Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	support
Kumar		
Age=18-24	−0.0494	223
Age=25-34	0.0365	1188
Age=35-44	0.0069	762
Age=45-54	0.0155	291
Age=55-64	−0.0650	130
Age=65+	−0.1752	17
Edu=Doctoral degree	0.0426	8
Edu=Professional degree	−0.2278	7
Edu=Master's degree	0.1159*	459
Edu=Bachelor's degree	0.0126	1193
Edu=Associate degree	−0.1270	176
Edu=College, no degree	−0.2258***	540
Edu=High School grad	−0.0739	122
Asian	0.0729	75
Black	0.2072***	356
Hispanic	0.0255	25
Multiracial	−0.1099	44
Other	0.1992	17
Unknown	−0.0246	7
White	−0.0436	1253
Gender=Female	−0.0370	2070
Gender=Male	0.0422	1974
Target=False	−0.0017	2199
Target=True	0.0329	1176
Parent=No	−0.0591***	1868
Parent=Prefer not to say	−0.1858	4
Parent=Yes	0.0205	2156
Trans=No	−0.0143	561
Trans=Prefer not to say	0.3697	7
Trans=Yes	0.5322**	31
Conservative	0.0552	1021
Independent	−0.0610	895
Liberal	−0.0419	1449
Other	−0.1127	21
Prefer not to say	−0.0148	54
Rel=No	−0.1069***	1069
Rel=Not very	−0.0706	241
Somewhat	0.0206	837
Very	0.0595	1199
Prefer not to say	−0.2144	13
False	0.0594	849
True	−0.0387	2051
SO=Bisexual	0.1639**	283
SO=Heterosexual	−0.0353	1170
SO=Homosexual	−0.0103	32
SO=Prefer not to say	−0.2791	23
Tech=Very negative	0.2157	4
Tech=Negative	−0.1614	176
Tech=Neutral	0.0705	320
Tech=Positive	−0.0283	1606
Tech=Very Positive	0.0263	1000
ToxProblem=Never	0.2975**	92
ToxProblem=Rarely	0.0024	498
ToxProblem=Occasionally	−0.0105	1096
ToxProblem=Frequently	−0.0181	958
ToxProblem=Very Freq	−0.0925	252

Table 5: Aposteriori unimodality results for the dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value.

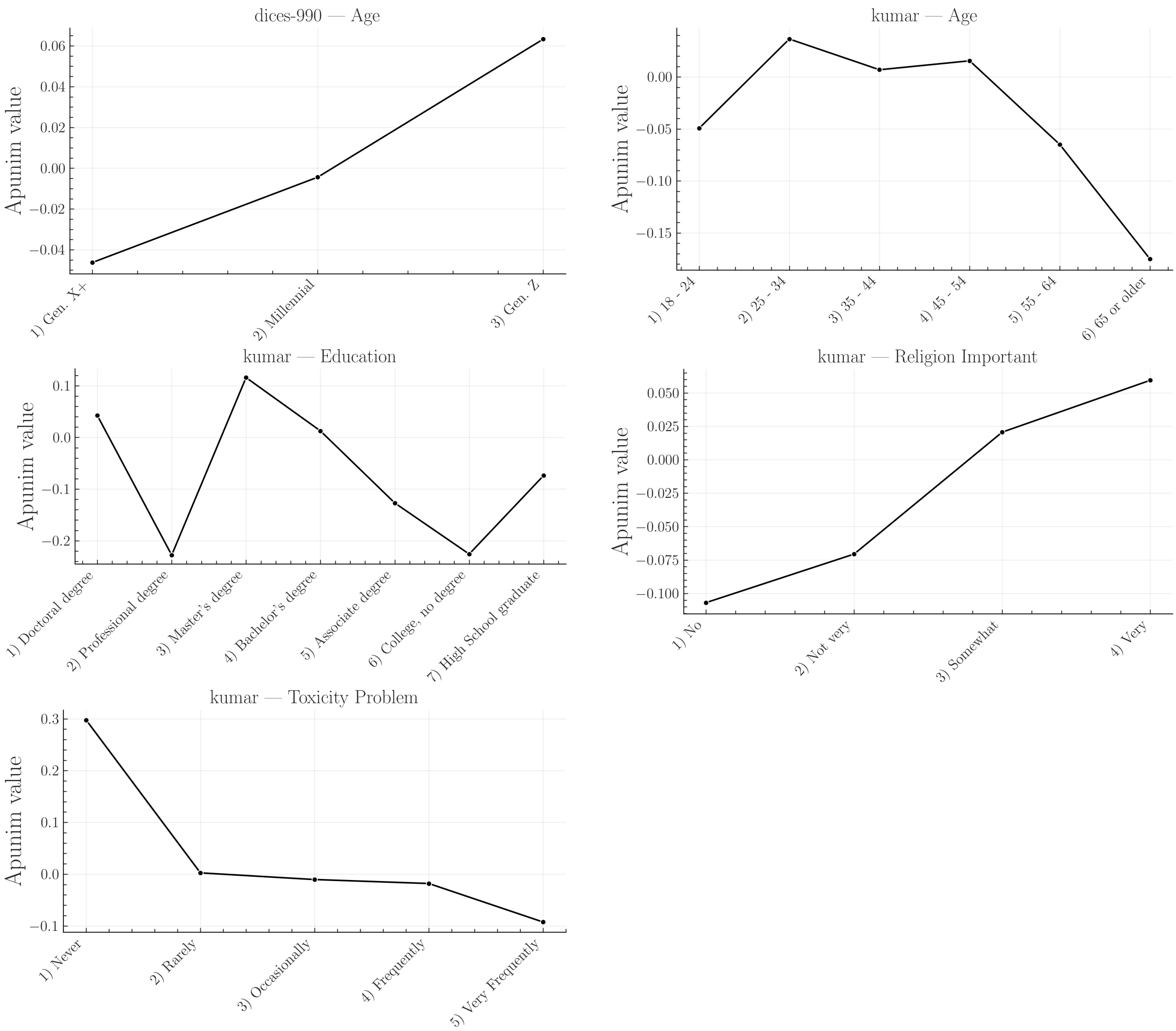

Figure 9: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal sociodemographic dimension. This figure is complementary to Figure 5. The values in the graphs were extracted from the Tables of F.

Group	apunim	support
DICES-350		
Race=AfricanAm.	-0.0428^{***}	9077
Race=Asian	0.0659^{***}	8138
Age=GenX+	0.0186	9703
Age=Millennial	−0.0059	11268
Age=GenZ	−0.0066	17528
Edu=College or higher	0.0124	23475
Edu=H. school or lower	−0.0152	12833
Edu=Other	-0.0214^{*}	2191
Gender=Man	−0.0193	19093
Gender=Woman	0.0215	19406
Race=Latino	−0.0030	6886
Race=Multiracial	−0.0001	5008
Race=White	−0.0198	9390

Table 6: Aposteriori unimodality results for the DICES-350 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.

Group	apunim	support
DICES-990		
Age=GenX+	0.0186	9703
Age=Millennial	−0.0044	38436
Age=GenZ	0.0633^{***}	13741
Edu=College or higher	0.0094	59142
Edu=H. school or lower	-0.0752^{***}	7419
Edu=Other	-0.0214^{*}	2191
Gender=Man	0.0275^{***}	32494
Gender=Woman	-0.0245^{***}	33471
Race=AfricanAm.	0.0911^{***}	5810
Race=Asian	0.0107	18428
Race=Latino	-0.1331^{***}	6610
Race=Multiracial	−0.0001	5008
Race=White	-0.1197^{***}	11392

Table 7: Aposteriori unimodality results for the DICES-990 Dataset. * denotes $p < 0.10$, ** denotes $p < 0.05$, and *** denotes $p < 0.01$. Rows shaded in gray indicate a negative apunim value. Groups for which no polarized comments among groups not included.